# Mahesh Marathe

Thane, Maharashtra, India

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## Professional Summary

Engineering student with a strong foundation in Python, SQL, and data analysis, with hands-on knowledge of NumPy, Pandas, Matplotlib, and Seaborn. Strong problem-solving and logical reasoning skills developed through DSA in Java, with knowledge of Data Mining, DBMS, MongoDB, Microsoft Excel, and AI/ML concepts. Interested in applying analytical and technical skills to solve business problems and derive meaningful insights from data.

## Education

**A.P. Shah Institute of Technology, Thane** Expected 2027
Bachelor of Engineering (B.E.) in Information Technology
Relevant Coursework: Data Structures and Algorithm, DBMS, Computer Networks and Network Analysis, Data Analytics, Machine Learning, AI, Web Development.

## Technical Skills

**Languages:** Python, Java , C, C++, JavaScript, HTML, CSS, SQL.
**Core CS:** Data Structures & Algorithms,Object-Oriented Programming (OOP).
**Data Analysis:** Pandas, NumPy, Matplotlib, Seaborn, Microsoft Excel.
**Data Mining :** Weka, RapidMiner.
**Databases:** DBMS, MongoDB.
**AI/ML :** Scikit-Learn (Random Forest, Xgboost ) , Pytorch ,(LSTM, NN).

## Projects

1. **PacketWatch — Network Traffic Analytics & Threat Detection** Sep 2026
   *Python*, *Pandas*, *NumPy*, *SQL/SQLite*, *Matplotlib*, *Django*
   - Built a network-traffic analytics pipeline that ingests PCAP/CSV data into normalized Pandas datasets and an SQLite analytical database.
   - Developed 10 rule-based detectors using SQL, Pandas and NumPy for identifying traffic patterns such as scanning, beaconing, DNS anomalies, brute-force attempts and data-exfiltration indicators.
   - Created automated visual reports and a Django dashboard with traffic trends, protocol distributions, top-talkers, DNS entropy and evidence tables.

2. **Streamline Task Management System** Jul 2025 – Oct 2025
   *React*, *Node.js*, *Express*, *MongoDB*
   - Built a role-based project management platform for managing **tasks, teams, approvals, and project reporting**.
   - Developed **REST APIs and MongoDB data models** for task assignment, workflow management, access control, and reporting.
   - Created **interactive dashboards** to track project and task metrics and provide actionable visibility into team performance.
   - Implemented **JWT authentication, Argon2id password hashing, 2FA, and audit logging** for secure application access.

## Certifications & Achievements

- **Smart India Hackathon (SIH 2026):** Shortlisted for the **Internal College Round**.
- **College-Level Hackathon:** Winner.
- Winner – College Coding Competition (Stock Prediction)
- **VishwaCTF:** Secured **13th rank among 300+ participating teams** in VishwaCTF, organized by CyberCell-VIIT.

## Additional Technical Experience

- Developed responsive web applications using **React** and reusable UI components.
- Integrated **MongoDB with backend services and REST APIs** for efficient data storage and retrieval.
- Worked with **Python/Django, Node.js/Express, and Linux**, with experience in debugging and managing application development workflows.